

Practical Assignment: Camera Calibration and Lens Undistortion

1. Assignment Overview

This assignment introduces a practical camera calibration workflow using a printed checkerboard. The goal is to understand how lens distortion affects images and how OpenCV can be used to estimate camera parameters and correct distortion.

2. Why This Assignment Matters

In Computer Vision and UAV applications, camera images are often used for mapping, navigation, object detection, landing pad detection, and visual odometry. Real cameras may introduce lens distortion, especially near the image borders. If this distortion is not corrected, geometric tasks such as corner detection, homography estimation, and pose estimation can become less accurate.

3. Dataset Preparation

Prepare the dataset using the following setup:

- Print a checkerboard pattern on A4 paper.
- Attach the printed checkerboard flat on a wall.
- Use an iPhone 14 Pro camera with the 1x lens only.
- Capture multiple images of the checkerboard from different angles and distances.
- Keep the checkerboard visible and clear in every image.

4. Assignment Requirements

Students are required to perform the following tasks:

- Collect a set of checkerboard images using the specified camera setup.
- Load the images in a Python notebook or script.
- Detect checkerboard corners using OpenCV.
- Estimate the camera matrix and distortion coefficients.
- Apply image undistortion to at least one test image.
- Compare the original and undistorted results.
- Write a short observation about how lens distortion affects the image.

5. Expected Output

The final submission should include:

- A folder containing the captured checkerboard images.
- A notebook or Python script for camera calibration.
- The estimated camera matrix and distortion coefficients.
- A visual comparison between the original and undistorted images.
- A short conclusion explaining the importance of camera calibration for UAV-based Computer Vision.

6. Success Criteria

The assignment is considered successful if the checkerboard corners can be detected in several images, the calibration parameters are generated, and the undistorted image shows a visible correction or a clear comparison with the original image.

Camera	iPhone 14 Pro, 1x lens
Calibration target	Printed A4 checkerboard attached flat on a wall
Main tool	OpenCV
Main result	Camera matrix, distortion coefficients, and undistorted image